Gaurav Singh

Varsity Lakes, Australia · hi@gvsh.cc · ORCID · GitHub · Google Scholar

Software developer and machine-learning researcher. First-author papers in the International Journal of Protective Structures (2024) and IOP Machine Learning (2026); Master's thesis compared diffusion, GAN, and VAE models on 14.7M air-quality observations. Currently researching transport decarbonisation.

Research interests

Machine learning for engineering problems; generative models for sparse and incomplete data; mathematical modelling and simulation; visual-mathematical literacy.

Education

2023–2025	Master of Information Technology Griffith University, Gold Coast, Australia; GPA 6.31/7 Thesis: “Enhancing Climate Forecasting with Advanced Generative Models for Synthetic AQI Data”, supervised by Dr. Elizabeth Chang.
2013–2015	Advanced Diploma, Innovation & Experience Design Srishti Institute of Art, Design and Technology, Bangalore, India
2008–2012	B.Tech, Computer Science & Engineering JNTU Anantapur, India; First Class with Distinction

Employment

2025–present	Software developer, Payments fintech, Australia Payment-processing development on a production platform: recurring billing, direct debit, and hosted payment pages.
2008–present	Founder and researcher, Mathscapes, India Independent studio for mathematics and algorithms; open-source libraries and mathematical design tools.
2018–2022	Design copilot, Topcoder, Remote Onboarded designers and ran crowdsourced challenges for Fortune 500 clients on the Topcoder platform.
2017–2022	Faculty member and researcher, Srishti Manipal Institute, Bangalore, India Head of Studies for the M.Des in Design Computation; founded and ran the Experimental Maths Lab.

Also: M56 (2015–2026), Griffith University (2022–2023), Art in Transit / Srishti (2016), Independent designer (2008–2015), S.Labs (2014); details in work.

Publications

A Framework for Visual-Mathematical Literacy in Applied Machine Learning: Why Representation Choices Shape What Models Can Learn. G. Singh, R. S. Dhari. IOP Machine Learning, 2026. doi:10.1088/3049-4761/ae7df3

Techno-economic pathways modeling and nonlinear optimized SEEA-ROI longitudinal dynamic simulation for decarbonising Australian heavy transportation systems. G. Singh, E. Chang, Y. Karaca. Fractals, 2026. doi:10.1142/S0218348X26400633

Automated detection of deformation mechanisms in re-entrant honeycomb auxetics using machine learning. G. Singh, R. S. Dhari, Z. Javanbakht. International Journal of Protective Structures, 2024. doi:10.1177/20414196241281069

ReRide: A Bike Area Network for Embodied Self-monitoring during Motorbike Commute. N. Bagalkot, G. Singh, V. Rath, T. Sokoler, A. Shukla. Proceedings of the Thirteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '19), 443-450, 2019. doi:10.1145/3294109.3300986

ReRide. N. Bagalkot, T. Sokoler, R. Shaikh, G. Singh, A. E. Lillie, P. Dixit, A. Rai, V. Chakravarthy, A. Senthil. Human-Computer Interaction - INTERACT 2017. Lecture Notes in Computer Science, vol 10516. Springer, Cham, 2017. doi:10.1007/978-3-319-68059-0_43

SnapTag: Leveraging Situated Memory to enhance self-efficacy for well-being. S. Baadkar, G. Singh, A. Saraf, N. Bagalkot. Proceedings of the India HCI 2014 Conference on Human-Computer Interaction, 136-141, 2014. doi:10.1145/2676702.2676719

Software

Iterflow: Composable Streaming Statistics for JavaScript. G. Singh. Mathscapes, 2026. doi:10.5281/zenodo.18610143

Teaching

30 courses and workshops in mathematics, computation, and human-computer interaction (2013–2022), mostly at Srishti Manipal Institute, Bangalore, India; the full list is in teaching.

Awards

2024	Griffith Award for Academic Excellence · Griffith University
2024	STEM Professional Gold Achievement · Griffith University
2023	Adobe Fund for Design · Design grant
2021	Adobe Fund for Design · Design grant
2017	Topcoder Open Design Champion · India Regionals
2012	Topcoder Open Studio Finals · 6th, Orlando
2011	Topcoder Open Studio Finals · 11th, Fort Lauderdale

Service

2018	Demonstrations track co-chair, IndiaHCI 2018 · Bangalore, India
2017	Cumulus Conference, organising team · Bangalore, India

Supervision

2020	Discovering Algorithmic Literacy through Explainable AI · D. Dixit, S. Singh
2020	Financial indicators that motivate impact investment · A. Avadhana, N. Patny, S. Mishra
2013	Location and arrival prediction · Shobha R., Pavani A., Kalpana

Skills

Machine learning: supervised learning, feature engineering, model evaluation and cross-validation; scikit-learn, PyTorch, TensorFlow/Keras.

Deep learning: convolutional networks for computer vision; transformers and attention; generative models: diffusion (DDPM), GANs (TimeGAN), VAEs (β-VAE).

Statistics & simulation: probability and statistical inference, exploratory data analysis, uncertainty quantification, nonlinear optimisation, techno-economic and longitudinal modelling.

LLMs & agents: LLM APIs and integration, structured extraction from documents, agentic workflows, evaluation.

Data & MLOps: Python (NumPy, Pandas), SQL; Matplotlib; Git, CI/CD; Docker, AWS.

Referees

Referees available on request.